Machine-checked algorithmic consequences of a $2^{o(n)}$ algorithm for #k-SAT

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Abstract

In October 2026 two groups claimed that the Exponential Time Hypothesis is false, and both formalized their algorithms in Lean. A refuted lower bound gives no algorithm by itself: each consequence needs a reduction to #SAT with few variables and, for a uniform bound, small clause width. We prove such consequences as Lean 4 theorems about Turing machines, on top of Anthropic's formalized $2^{o(n)}$ algorithm, using only the standard axioms. The results include: counting the satisfying assignments of CNFs of any width with up to $n \cdot 2^{(\lg n)^{1/4}/(\lg\lg n)^2}$ clauses; counting k-colorings for $\lg k \le (\lg n)^{1/6-\theta}$ with one machine; the Max-SAT spectrum; the density of states of the q-state Potts model and the Tutte polynomial on the hyperbolas $(x-1)(y-1) = q \in \mathbb{N}$ for graphs with $O(n)$ edges; the parity of the number of Hamiltonian cycles in graphs with $n(\lg n)^{1/2-\theta}$ edges; the parity of connected solutions (Steiner trees, connected vertex covers) on dense graphs; randomized algorithms with one-sided error for Hamiltonian cycle on $O(n)$-edge graphs and for Steiner tree and connected vertex cover on any graph, via a formalized isolation lemma for arbitrary sets; and Vertex Cover in $2^{\varepsilon k}\operatorname{poly}(n)$ for every $\varepsilon > 0$ without a kernel. Each running time is $2^{\varepsilon n}$ (or $2^{\varepsilon k}\operatorname{poly}(n)$) for every $\varepsilon > 0$. The Hamiltonian cycle count is checked against Mathlib's definition. Everything is conditional only on the correctness of Lean and of the statements of Anthropic's formalization; it is not conditional on any unproved hypothesis.

1 Introduction

The Exponential Time Hypothesis (ETH) of Impagliazzo and Paturi [4] says that 3-SAT on n variables needs time $2^{\Omega(n)}$. Much of fine-grained complexity rests on it. In October 2026 two groups claimed that it is false.

- OpenAI [1] claims #k-SAT for every fixed k in time $(n+m)^{O_k(1)} \cdot 2^{O_k(n \lg\lg n/\sqrt{\lg n})}$, deterministic and in polynomial space. The core tool is the contraction of Boolean tensor networks with s wires in time $2^{O(s/\sqrt{\log s})}$.
- Anthropic [2] claims #k-SAT with one machine for all k in time $2^{C\min(k^2 n \lg\lg n/\sqrt{\lg n},\, km/\sqrt{\lg m})}$ with an absolute constant C, via sum-product computations on the clause graph.

Both papers are unrefereed and both come with Lean formalizations [3]. A review led by Impagliazzo, Paturi and Williams is under way; its organizers stress that taking part is not a certification. If the claims hold, ETH, SETH, #ETH, Gap-ETH and NSETH fall. The OV conjecture, the CNF-SAT hypothesis for $\operatorname{poly}(n)$ clauses of unbounded width, the k-Clique conjecture and the Set Cover conjecture still stand. The algorithms are galactic: Anthropic reports no gain over 2^n below $n \approx 10^{10}$.

	Statement (time $2^{\varepsilon n}$ resp. $C2^{\delta n}$ for every $\varepsilon, \delta > 0$)	Status
A1	#CNF-SAT, any width, $\leq Dn$ clauses	known [6] + ¬ETH
A1′	#CNF-SAT, any width, $m \leq n\, 2^{(\lg n)^{1/4}/(\lg\lg n)^2}$, one machine	new
A2	#k-colorings for all k with $\lg k \leq (\lg n)^{1/6-\theta}$, one machine	growing k new
N5	#homomorphisms $G \to H$, H in the input, $\lg \lvert V(H)\rvert \leq (\lg n)^{1/6-\theta}$, one machine	generalizes A2
A4	Max-SAT spectrum: #assignments satisfying exactly t clauses, width r, $\leq Dn$ clauses	new
N3	Potts density of states, q colors, $\leq Dn$ edges; Tutte polynomial on $(x-1)(y-1) = q$	corollary
N6	0/1 permanent in $C(n+2)^C 2^{\varepsilon m}$, m = number of ones	corollary
A5a	parity of #Hamiltonian cycles, degree $\leq \Delta$	new
A5a′	parity of #Hamiltonian cycles, $\leq n(\lg n)^{1/2-\theta}$ edges, any degree, one machine	new
N2	parity of #connected vertex sets of size ℓ satisfying a CNF; any density	new
A5b	Hamiltonian cycle, one-sided error, $\leq Dn$ edges	new
N2′	Steiner tree, connected vertex cover, one-sided error, any density	new
A3	Vertex Cover in $C(n+2)^C 2^{\varepsilon k}$	known [14] + ¬ETH
A3′	Nonblocker in $C(n+2)^C 2^{\varepsilon k}$	known [12, 13] + ¬ETH

Table 1: The results. A1, A3 and A3′ are formalizations of known equivalences, N3, N5 and N6 are direct corollaries; the others are our derivations.

Corollaries are not automatic. A lower bound that is refuted gives no algorithm. To get a $2^{o(n)}$ algorithm for a problem X one needs a reduction from X to (#)SAT with $o(n\sqrt{\lg n}/\lg\lg n)$ variables and, for the uniform bound, small clause width. Problems in size-constrained SNP reduce this way [5]. Problems with global constraints (connectivity, orderings, large numbers) and dense pairwise interactions do not. Our results are reductions of this kind, some standard and some new, with two recurring tools: Cut & Count [7] turns connectivity into a parity of a count, and sparse parity checks [8] isolate one solution without new variables per solution.

What we prove and how. Every result is a Lean 4 theorem that a single Turing machine (Mathlib's multi-stack model TM2, finite alphabets) halts on a stated input with a stated output within a stated number of steps. The theorems are built on Anthropic's Lean development, which proves its $2^{o(n)}$ algorithm as a theorem; so our results are unconditional theorems of Lean, as sound as Anthropic's formalization. They use the axioms `propext`, `Classical.choice` and `Quot.sound` only. The development has about 11,000 lines in 33 files and 50 main theorems, whose axioms are printed by a check file. Source: `https://github.com/ssenge/eth-corollaries`.

Lean cannot check that a formal statement says what is intended. The definitions behind the statements must be read by hand: ours (`numHom` and `hmat`, checked against Mathlib's $G \to_g H$, `numHamCycles`, `spectrum`, `pottsSpec`, `mono`, `comps`, `tutte`, `Seed`, the input codes) and Anthropic's (the TM2 model and their statement files). For one of ours, the number of Hamiltonian cycles, we prove agreement with Mathlib's independent definition (Section 7).

Results. Table 1 lists the results. "One machine" means one fixed TM; constants C and thresholds n_0 depend on the stated parameters only. We have not finished a literature check for the entries marked "new".

2 The interface

We use three parts of Anthropic's development as black boxes.

Theorem 1 (SIZED, Anthropic)**.** *For every width w and $\delta > 0$ there are a machine SIZED_w (a polynomial-time stepped computation) and c such that for all N, $s \le N$ and every CNF ψ on N variables with clauses of at most w literals, SIZED_w on $(1^N, s, \psi)$ returns the number $\mathrm{cnt}_s(\psi)$ of satisfying assignments with exactly s true variables, within $c{\cdot}2^{\delta N}$ steps of size at most $c(N+|\psi|+2)^c$.*

The number of clauses of ψ only enters the polynomial factor. This is what makes dense graphs possible whenever the variables are the vertices.

Stepped machines, trees and the host. Anthropic's machines are *stepped*: a computation is cut into polynomial-time steps, and a theorem turns a stepped machine that runs in T steps of state size S into a TM2 machine with time $q(T+1)(|x|+S+1)^q$. A depth-first walk over the tree of a step function $\sigma \to \mathrm{Option}(\sigma \times \sigma)$ that sums a stepped computation over the leaves is again stepped (`walkSum`), with a bound on the number of leaves when every path has at most D steps and at most T/q "heavy" steps. The host theorem `tm_of_stepped` gives time $\lfloor C2^{\delta n} \rfloor$ from $c\,2^{\delta n/2}$ steps; we add an FPT version `tm_fpt` with time $\lfloor C(n+2)^C 2^{\delta p} \rfloor$ for a parameter p.

Encodings. A graph on n vertices is coded as 1^n0 followed by its adjacency matrix, a number t in unary as 1^t0, a CNF as 1^n0 followed by its clauses, each literal in $\lfloor \lg 2n \rfloor + 1$ bits.

3 Counting

3.1 CNF of any width (A1, A1′)

Theorem 2 (`cnf_counts`)**.** *For every D and $\delta > 0$ one machine counts the satisfying assignments of every CNF with n variables and at most Dn clauses, of any width, within $C2^{\delta n}$ steps.*

This is the width–density duality of Calabro, Impagliazzo and Paturi [6] combined with ¬ETH. The literal (u, b) becomes the variable $2u$ or $2u+1$, with pair clauses making them complementary, so every clause is positive. Anthropic's clause-shortening tree from their Dominating Set corollary cuts clauses to width w; we replace its invariant (which needs at most n clauses) by one for $M = Dn$ clauses. The tree has at most $2^{(2/w)\lg(ew(D+2))n}$ leaves, and SIZED counts at each leaf.

Theorem 3 (`cnf_grow`)**.** *One machine: for every $\delta > 0$ there is n_0 such that for every CNF on $n \ge n_0$ variables with at most $n \cdot 2^{(\lg n)^{1/4}/(\lg\lg n)^2}$ clauses of any width, it outputs the number of satisfying assignments within $2^{\delta n}$ steps.*

The width is now read off the node: with $\ell = \lfloor \lg 2n \rfloor$ and $L = \lfloor \lg \ell \rfloor$, $w = 2^{\lfloor L/4 \rfloor - \lfloor \lg L \rfloor} + 1 \approx \ell^{1/4}/L$. The leaves are counted by Anthropic's uniform machine, whose exponent $Cw^2 N \lg\lg N/\sqrt{\lg N}$ is at most $32CN/L = o(n)$ since $w^2L^2 \le 16\sqrt{\ell}$. The tree coefficient is at most $(8 + 6\sqrt{w})/w + 8/(L-1) \to 0$ for $\lg D \le 4w/(L-1)$. The method stops at density $n2^{o((\lg n)^{1/4}/\sqrt{\lg\lg n})}$.

3.2 Colorings with a growing number of colors (A2)

Theorem 4 (`colorings_subexp`)**.** *One machine counts the proper k-colorings of an n-vertex graph, given k and the graph, within $2^{\varepsilon n}$ steps, for every $\varepsilon > 0$, $0 < \theta < 1/6$, $n \ge n_0(\theta, \varepsilon)$ and all $2 \le k$ with $\lg k \le (\lg n)^{1/6-\theta}$.*

Colors are written in $b = \lceil \lg k \rceil$ bits; the CNF has $nb + 4$ variables, width $2b$, and one satisfying assignment per proper coloring. The uniform bound with width $2b$ and $N = nb$ variables is $O(n(\lg n)^{-2\theta})$ in the exponent.

Homomorphisms (N5). The same encoding counts homomorphisms into a target that is part of the input. Let H be a relation on $\{0, \dots, h-1\}$ (loops allowed, not necessarily symmetric), given by its h^2 entries, and let $\hom(G, H)$ be the number of maps $f : V(G) \to [h]$ with $H(f(u), f(v))$ for every edge uv.

Theorem 5 (`hom_subexp`, `hom_subexp_graph`)**.** *One machine, given h in unary, the matrix of H and an n-vertex graph G, outputs $\hom(G, H)$ within $2^{\varepsilon n}$ steps, for every $\varepsilon > 0$, $0 < \theta < 1/6$, $n \geq n_0(\theta, \varepsilon)$ and all $h \geq 2$ with $\lg h \leq (\lg n)^{1/6-\theta}$. For a simple graph H the output is the number of Mathlib homomorphisms $G \to_g H$ (`numHom_simple`).*

With $b = \lceil \lg h \rceil$ bits per vertex, every edge uv of G and every pair (a, c) with $H(a, c)$ false give the clause "not both $u \mapsto a$ and $v \mapsto c$" of width $2b$ (`homB`, `cntN_homB`). The variables and the width are those of A2; only the number of clauses grows, to at most $n^2h^2 + O(nh)$, which the uniform bound does not see beyond a polynomial factor. For $H = K_h$ this is A2 (`numColorings_eq`); for H an edge with a loop at one end it counts independent sets. Under ETH, deciding whether $G \to H$ exists takes $|V(H)|^{\Omega(|V(G)|)}$ time in general [11].

3.3 Spectra (A4) and the Potts model (N3)

Theorem 6 (`maxsat_spectrum`)**.** *For every width r, density D and $\delta > 0$, one machine outputs, given t and a CNF φ with clauses of at most r literals and at most Dn clauses, the number of assignments that satisfy exactly t clauses of φ, within $C2^{\delta n}$ steps.*

On $2n + m$ variables (x_u, $\bar{x}_u$, and $y_j \leftrightarrow C_j$) the satisfying assignments with exactly $n + t$ true variables are the assignments with exactly t satisfied clauses. This covers the Max-Cut spectrum and the density of states of the ± 1 Ising model on $O(n)$-edge graphs.

For a graph G and q colors let N_j be the number of colorings $\sigma : V \to [q]$ with exactly j monochromatic edges, $k(A)$ the number of components of (V, A), and $T(G; x, y) = \sum_{A \subseteq E} (x-1)^{k(A)-k(E)}(y-1)^{|A|+k(A)-n}$ the Tutte polynomial.

Theorem 7 (`potts_spectrum`, `potts_fk`, `tutte_hyperbola`, `tutte_tm`)**.** *For every $q \geq 1$, D and $\delta > 0$, one machine outputs N_j given G with at most Dn edges and j, within $C2^{\delta n}$ steps. Moreover, in every commutative ring, $\sum_j N_j(1+v)^j = \sum_{A \subseteq E} q^{k(A)} v^{|A|}$, and in every field, if $(x-1)(y-1) = q$ and $x, y \neq 1$, then $(x-1)^{k(E)}(y-1)^n T(G; x, y) = \sum_j N_j y^j$.*

The CNF has variables x_{vq+c} (v has color c) and y_i (edge i is monochromatic), $nq + m$ in all, and width $q + 3$. Its satisfying assignments with $n + j$ true variables are the colorings with j monochromatic edges. The identities are the Fortuin–Kasteleyn expansion [9]: expand $\prod_e (1 + v[e \text{ mono}])$ over $A \subseteq E$ and count the colorings constant on the components of (V, A), which are $q^{k(A)}$ many. This gives the chromatic polynomial at q (the point $(1-q, 0)$) and the flow polynomial at q (the point $(0, 1-q)$). Under #ETH, Dell et al. [10] prove lower bounds of the form $2^{\Omega(n/\operatorname{polylog} n)}$ for such evaluations on sparse graphs.

3.4 The 0/1 permanent (N6)

Theorem 8 (`perm_fpt`). *For every $\varepsilon > 0$, one machine, given an $n \times n$ matrix A with entries in $\{0,1\}$ (n in unary, then the n^2 entries), outputs Mathlib's `Matrix.permanent` of A within $C(n+2)^C 2^{\varepsilon m}$ steps, where m is the number of ones of A.*

There is one variable per one of A, and a clause $\neg x_a \vee \neg x_b$ for every two ones in the same row or column (`permF`, width 2, at most m^2 clauses). An assignment with exactly n true variables chooses n ones, no two in a column, hence one in every column, and their rows are distinct; so it is a permutation σ with $A_{\sigma(i),i} = 1$ for all i, and conversely (`cntS_permF`). These are the nonzero terms of the permanent (`permanent_eq`). SIZED counts the solutions of size n in $c\,2^{\varepsilon m}$ steps; the number of clauses does not enter the exponent. Under #ETH, the 0/1 permanent with m nonzero entries has no $\exp(o(m/\log n))$ algorithm [10, Thm. 1.3]; with the claimed refutation it takes $2^{\varepsilon m}\,\mathrm{poly}(n)$ for every $\varepsilon > 0$.

4 Connectivity through parity

4.1 Hamiltonian cycles (A5a, A5a′)

Let $\mathrm{hc}(G)$ be the number of connected spanning subgraphs of G with all degrees 2.

Theorem 9 (`hc_parity_subexp`, `hc_parity_sparse`). *For every Δ, one machine decides whether a graph with all degrees at most Δ has an odd number of Hamiltonian cycles, within $2^{\varepsilon n}$ steps for $n \geq n_0(\varepsilon)$. One machine decides it for all graphs with at most $n(\lg n)^{1/2-\theta}$ edges, within $2^{\varepsilon n}$ steps for $n \geq n_0(\theta, \varepsilon)$.*

The Cut & Count CNF [7] has an edge bit per edge and a side bit per vertex, and says that every vertex is on exactly two chosen edges, the ends of a chosen edge share a side, and vertex 0 has side 0. A 2-factor with c cycles has 2^{c-1} side maps, which is odd exactly when $c = 1$. For bounded degree the formula has width $\Delta + 3$; for any degree a sequential counter with two bits per (edge, end) replaces the cardinality constraints, giving width 3 and $5m + n + 2$ variables. Anthropic's bound $2^{9cN \lg\lg N/\sqrt{\lg N}}$ is then $2^{o(n)}$ for $m = o(n\sqrt{\lg n}/\lg\lg n)$.

4.2 Connected solutions on dense graphs (N2)

Theorem 10 (`conn_parity_tm`, `cvc_parity_tm`). *For every width r, exponent D and $\delta > 0$ one machine, given a graph G on n vertices of any density, $1 \leq \ell \leq n$ and a CNF φ on the vertices with clauses of at most r literals and at most $(n+2)^D$ clauses, outputs the parity of the number of $X \subseteq V$ with $|X| = \ell$, $X \models \varphi$ and $G[X]$ connected, within $C2^{\delta n}$ steps.*

The variables are a vertex bit x_v, a side bit s_v and a complement bit $\bar{s}_v$; the clauses are those of φ, $\bar{s}_v = \neg s_v$, $s_v \to x_v$ and $s_u = s_v$ on edges inside X. The satisfying assignments with $n + \ell$ true variables are the pairs (X, s), $2^{\mathrm{cc}(G[X])}$ per X, and half their number is odd exactly when $G[X]$ is connected. The formula has $3n$ variables whatever the number of edges. With φ the unit clauses of a terminal set this counts the vertex sets of Steiner trees on ℓ vertices; with φ the edge clauses, connected vertex covers.

5 Isolation and one-sided error

Theorem 11 (`isolation`; Calabro, Impagliazzo, Kabanets, Paturi [8]). *For every $\delta > 0$ there are k, t such that for $M \geq M_0(\delta)$ and every nonempty $S \subseteq \{0,1\}^M$, at least a $2^{-\delta M}$ fraction of the*

seeds leaves exactly one point of S. A seed is $3M+3$ parity checks on k coordinates, a number of them to use, and for every coordinate a value in $\{0,\dots,t-1\}$ and a bit (the coordinate is fixed to the bit when the value is 0).

[8] states this for solution sets of k-CNFs, but the proof uses only that S is nonempty. A check on k coordinates becomes a chain of k auxiliary variables $a_l = a_{l-1} \oplus (c_l \wedge x_{j_l})$ in clauses of width at most 3, determined by the main variables; so isolation keeps the number of variables linear, unlike Valiant–Vazirani hashing.

Theorem 12 (`ham_randomized`, `conn_randomized`)**.** *(a) For every $D, \delta, \varepsilon > 0$ there are k, t, n_0 and one machine that, on a graph with $n \geq n_0$ vertices and at most Dn edges and a seed ω, halts within $2^{\varepsilon n}$ steps with a bit that is 0 for every seed if G has no Hamiltonian cycle, and 1 for at least a $2^{-\delta n}$ fraction of the seeds if it has one. (b) The same, for graphs of any density, holds for the existence of a vertex set of size ℓ that satisfies a CNF φ as in N2 and induces a connected subgraph; in particular for Steiner trees on ℓ vertices and connected vertex covers of size ℓ, within $C2^{\varepsilon n}$ steps.*

For (a), the checks act on the edge bits; the count modulo 2 of the Hamiltonian cycles passing the checks is odd for a $2^{-\delta m}$ fraction of seeds. For (b), the checks act on the vertex bits, and every chain bit gets a complement bit so that the number of true variables stays fixed, as SIZED requires. The machines read the seed from the input, so one machine serves every δ. Repeating with $2^{2\delta n}$ independent seeds gives error $e^{-2^{\delta n}}$; this step is not formalized.

6 Parameterized by the solution size: Vertex Cover (A3), Nonblocker (A3′)

Theorem 13 (`vc_fpt`)**.** *For every $\varepsilon > 0$, one machine, given a graph G on n vertices (of any density) and $k \leq n$, decides whether G has a vertex cover of at most k vertices within $C(n+2)^C 2^{\varepsilon k}$ steps.*

Cai and Juedes [14] showed that such an algorithm refutes ETH; the converse with a linear kernel is folklore. We avoid the kernel. While the budget k' is positive and a vertex v has at least d distinct neighbours, branch: v is in the cover (budget $k'-1$) or all of $N(v)$ is (budget $k' - |N(v)| \leq k' - d$). A node is good iff one child is (`good_step`). Along a path the budget drops by 1 or by at least d, so with Anthropic's tree bound the tree has at most $\sum_{j \leq (k+d)/d} \binom{k+d}{j} \leq 2^{\varepsilon'(k+d)}$ leaves once $(1/d)\lg(ed) \leq \varepsilon'$ (`leaves_le`). At a leaf all degrees are below d, so a cover of size k' leaves at most $k'd$ vertices on the edges (`card_V_le`); SIZED decides the leaf on these vertices, in $2^{\delta kd}$ steps, and $\delta = \varepsilon/(4d)$ makes it $2^{\varepsilon k/4}$.

Nonblocker. The dual of Dominating Set asks for k vertices whose complement dominates.

Theorem 14 (`nb_fpt`)**.** *For every $\varepsilon > 0$, one machine, given G on n vertices and $k \leq n$, decides whether G has a dominating set of at most $n-k$ vertices within $C(n+2)^C 2^{\varepsilon k}$ steps.*

Isolated vertices lie in every dominating set, so the question is the same for the graph induced on the n' vertices with a neighbour and the bound $n' - k$ (`nb_iff`). That graph has no isolated vertex, and by Ore's theorem [12] it has a dominating set of at most $n'/2$ vertices: the complement of a minimum dominating set dominates (`ore`). So the answer is yes for $2k \leq n'$, no for $n' < k$, and otherwise $n' < 2k$ and Anthropic's counter of dominating sets of a given size runs on the induced graph in $c\,2^{(\varepsilon/2)n'} \leq c\,2^{\varepsilon k}$ steps. This is the linear kernel of [13] in its simplest form, with the claimed refutation as the exponential part.

7 Checking a definition against Mathlib

Mathlib defines Hamiltonian cycles as closed walks that are cycles and visit every vertex once (`Walk.IsHamiltonianCycle`), and `SimpleGraph.IsHamiltonian`.

Theorem 15 (`numHamCycles_pos_iff`, `numHamCycles_eq`)**.** *For $n \neq 1$, $\mathrm{hc}(G) > 0$ iff G is Hamiltonian in Mathlib's sense. Moreover $\mathrm{hc}(G)$ equals the number of edge sets of Mathlib Hamiltonian cycles of G.*

The proof uses Mathlib's result that a connected graph with all degrees 2 is a single cycle through its vertices. With it, (a) of the previous section is stated directly for `G.IsHamiltonian` (`ham_randomized_mathlib`).

8 Discussion

The results mark where the claimed algorithms reach. The variable count is the currency: $O(n)$ variables give $2^{\varepsilon n}$, $n \cdot r(n)$ variables with $r = o(\sqrt{\lg n}/\lg\lg n)$ still give $2^{o(n)}$, and the clause width must stay below about $(\lg n)^{1/4}$ for the uniform bound. Open directions: deterministic decision versions of A5b and N2′; Hamiltonian cycle and Steiner tree with more edges; the Tutte polynomial off the integer hyperbolas; Nonblocker and other linear-kernel problems; and the Tier B problems of the repository notes (dense Max-Cut, the permanent, CNF-SAT with $\mathrm{poly}(n)$ clauses, planar problems below $2^{O(\sqrt{n})}$, Subset Sum), where the refutations remove a lower bound but no reduction is known.

Use of AI

Claude Opus 5.5 (Anthropic) was used throughout this research: to explore approaches, to develop and check proofs, to write the Lean formalization, and to draft the text.

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